Situation-Aware Player Similarity via Trait–State Decomposition on Heterogeneous Graphs

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Abstract

We study substitute-oriented soccer player retrieval: given a query player, find those who would make similar on-ball decisions under the same tactical state. We model each possession as a heterogeneous graph over events and 360 freeze-frame players, learn contextual state embeddings with a GATv2 encoder, extract global trait vectors via attention pooling, and condition action prediction through FiLM modulation. A systematic ablation across 11 pipeline variants examines eight modular components, with three heuristic baselines used as supplementary reference points.

Four main observations emerge from the tested variants. First, in this study, position ablation combined with split-context edges and stronger uniformity settings yields the most stable non-collapsed embeddings and the strongest cross-context consistency among the configurations evaluated. Second, player-aware sampling appears to optimize player re-identification more than cross-player substitute discovery. Third, a coarse empirical behavioral (EB) metric indicates that the acts-in-dropout family produces stronger observed behavioral matching than random same-position peers, although the EB protocol is too coarse to cleanly isolate individual component effects. Fourth, under the chosen weighted composite, `acts_in_dropout_pos_gu` attains the highest overall score (0.867), reflecting the strongest overall balance across behavioral fidelity, cross-context robustness, and external validation rather than dominance on every individual metric.

1 Introduction

The goal is to retrieve soccer players who function as plausible *substitutes* for a query player: not merely similar in season-level production, but similar in how they would decide under the same tactical situation. Aggregate statistics confound player behavior with team quality, role assignment, and tactical scheme. A fullback on a dominant side and one on a transitional side may post very different numbers even if their decision-making under matched states is identical.

We formalize this with a behavioral discrepancy:

$$\Delta(p,q) = \mathbb{E}_{s\sim\mathcal{D}_S}\Big[\mathrm{JS}\big(P_\theta(\cdot \mid s, z_p),\ P_\theta(\cdot \mid s, z_q)\big)\Big],$$

where s is a masked pre-action game state, $z_p \in \mathbb{R}^d$ is a learned trait embedding, and P_θ is the conditional action policy. The practical goal is to learn z_p so that cosine proximity approximates low $\Delta(p,q)$.

Section 3 describes data and state representation. Section 4 presents the model architecture. Sections 5–6 detail the training objective and the eight modular components studied. Sections 7–8 present evaluation and results, followed by model selection (Section 9), discussion (Section 10), and conclusion (Section 11).

2 Related Work

Conventional player comparison relies on aggregated statistics or composite ratings. Decroos et al. [3] (VAEP) and Fernández et al. [4] value individual actions and possessions but do not yield continuous player embeddings suitable for retrieval. GNNs have been applied to team-sport analysis: Yeh et al. [11] model basketball trajectories as spatio-temporal graphs with graph neural networks and variational RNNs, enabling counterfactual trajectory prediction. Our architecture draws on GATv2 [2], FiLM [8], InfoNCE [7], focal loss [6], and alignment and uniformity analysis on the hypersphere [10]. Our contribution is integrative: the individual components are established, but their combination for behavioral player similarity, and the systematic ablation revealing which interactions matter, is novel in this domain.

3 Data and State Representation

The system uses StatsBomb 360 open data [9], which augments standard event data with freeze-frame snapshots of all visible players at each on-ball action. The working corpus contains approximately 51,000 possessions and 737,000 events across 323 matches from 7 competition-seasons with 360 coverage. Players with fewer than 50 possessions are excluded, yielding 1,633 embedded players (311 embedded players appear in $\geq$2 competitions).

Each event is encoded as a 126-dimensional vector mixing categorical fields (event type, play pattern, position, body part, action/outcome descriptors), continuous fields (pitch coordinates, displacement, pass geometry, pressure), and contextual fields (pitch zone, match period, spatial-360 summaries from freeze frames). The spatial-360 block is zeroed in the graph representation so the GNN must learn spatial context from explicit player nodes.

Events are grouped by `possession_number` into possession sequences with per-event timestamps. Possessions with fewer than 2 or more than 200 events are discarded. Possession-level labels (ends-in-shot, ends-in-goal, total xG) provide a secondary supervisory signal.

4 Architecture

The architecture decomposes each prediction into a *state pathway* (what the current situation affords) and a *trait pathway* (how a specific player tends to act). Figure 1 shows the graph topology; Figure 2 illustrates the full pipeline.

Heterogeneous graph. Each possession is encoded as a heterogeneous graph with two node types: **event nodes** (masked 126-D features) and **player nodes** ([position_idx, team_flag, dx, dy], covering both the on-ball actor and off-ball players from the 360 freeze frame). Edges encode temporal adjacency (with normalized time-delta), actor–event links (`acts_in` / `performed_by`), and context edges connecting off-ball players to their focal event. When split-context mode is active (Section 6), context edges separate into teammate and opponent relations.

GNN encoder. A two-layer heterogeneous GATv2 [2] encoder (`HeteroConv` [5]) processes the graph: Layer 1 maps 64-D inputs to 512-D (4 heads $\times$ 128-D); Layer 2 projects back to 64-D (single head), with learned skip connections.

Strict masking. To prevent label leakage, we zero out event type (the prediction target), end location, displacement, position, body part, action/outcome descriptors, and action-specific scalars. Unmasked features preserve ball location, play pattern, pressure, pitch zone, and match period.

Global trait extraction. A player appearing across many possessions accumulates per-possession embeddings, aggregated via learned attention pooling [1]:

$$z_p = \sum_i \alpha_{p,i}\, h_{p,i}^{\text{player}}, \qquad \alpha_{p,i} = \text{softmax}_i(g(h_{p,i}^{\text{player}})),$$

where g is a two-layer scoring network.

FiLM conditioning. Prediction uses Feature-wise Linear Modulation [8]:

$$\widetilde{h}^{\text{event}}(p) = \big(1 + \gamma(c)\big) \odot h^{\text{event}} + \beta(c),$$

where $c = [z_p;\ r_p]$ and r_p is an optional 16-D positional embedding. Prediction heads output logits for action type (14-way), angle bin (9-way), and length bin (5-way).

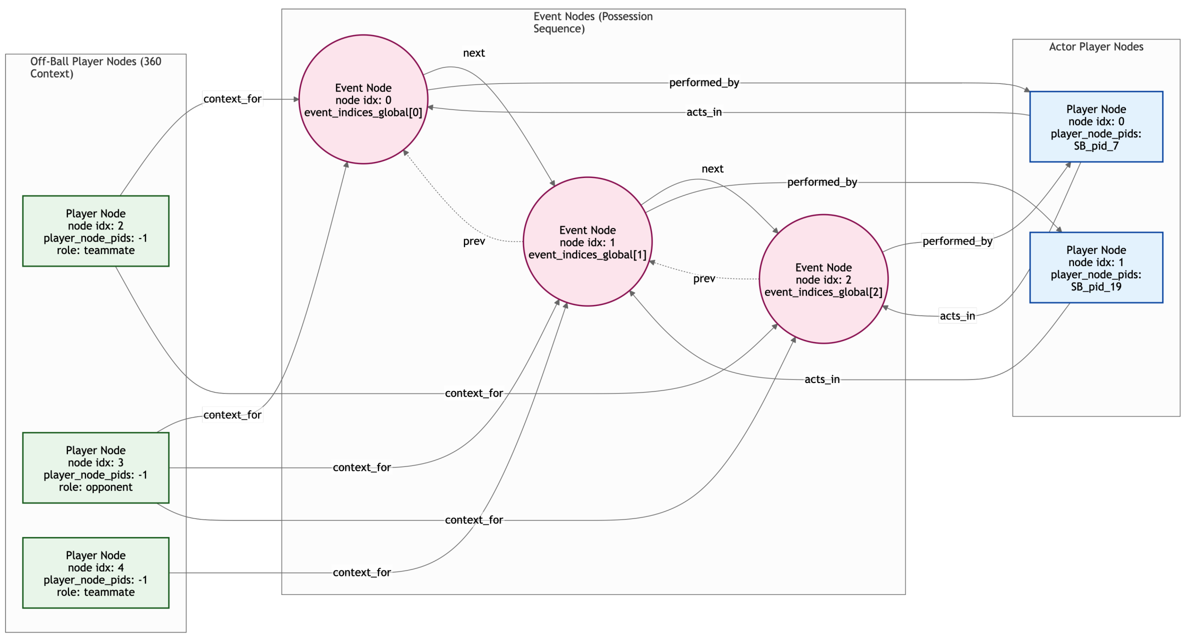


Figure 1: Heterogeneous graph topology for a single possession. Pink circles are event nodes linked by temporal edges; blue rectangles are actor player nodes with real player IDs connected via `acts_in`/`performed_by`; green rectangles are off-ball player nodes from 360 freeze frames connected via `context_for` edges.

5 Training Objective

The total loss combines supervised prediction with embedding-geometry regularization:

$$\mathcal{L} = \mathcal{L}_{\text{action}} + \lambda_{\text{out}}\,\mathcal{L}_{\text{outcome}} + \lambda_{\text{con}}\,\mathcal{L}_{\text{contrast}} + \lambda_{\text{uni}}\,\mathcal{L}_{\text{uniformity}} + \lambda_{\text{align}}\,\mathcal{L}_{\text{align}} + \lambda_{\text{pos}}\,\mathcal{L}_{\text{pos}}.$$

$\mathcal{L}_{\text{action}}$ uses focal loss [6] over action type, angle, and length bins with class-rebalancing weights. $\mathcal{L}_{\text{outcome}}$ is BCE for shot/goal prediction from h^{event} alone. $\mathcal{L}_{\text{contrast}}$ is InfoNCE [7] on actor embeddings with same-player positives and same-position-group hard negatives. $\mathcal{L}_{\text{uniformity}}$ is the Gaussian-potential loss [10] on L2-normalized z_p, optionally with group-weighted repulsion. $\mathcal{L}_{\text{align}}$ provides cross-batch alignment via EMA memory bank. $\mathcal{L}_{\text{pos}}$ predicts coarse position group from z_p. Default weights: $\lambda_{\text{out}} = 0.5$, $\lambda_{\text{con}} = 0.5$, $\lambda_{\text{uni}} = 0.3$, others inactive. Training uses Adam ($\text{lr} = 10^{-3}$, weight decay 10^{-5}), gradient clipping at 1.0, batch size 96, 70/15/15 match-level splits, and ReduceLROnPlateau scheduling.

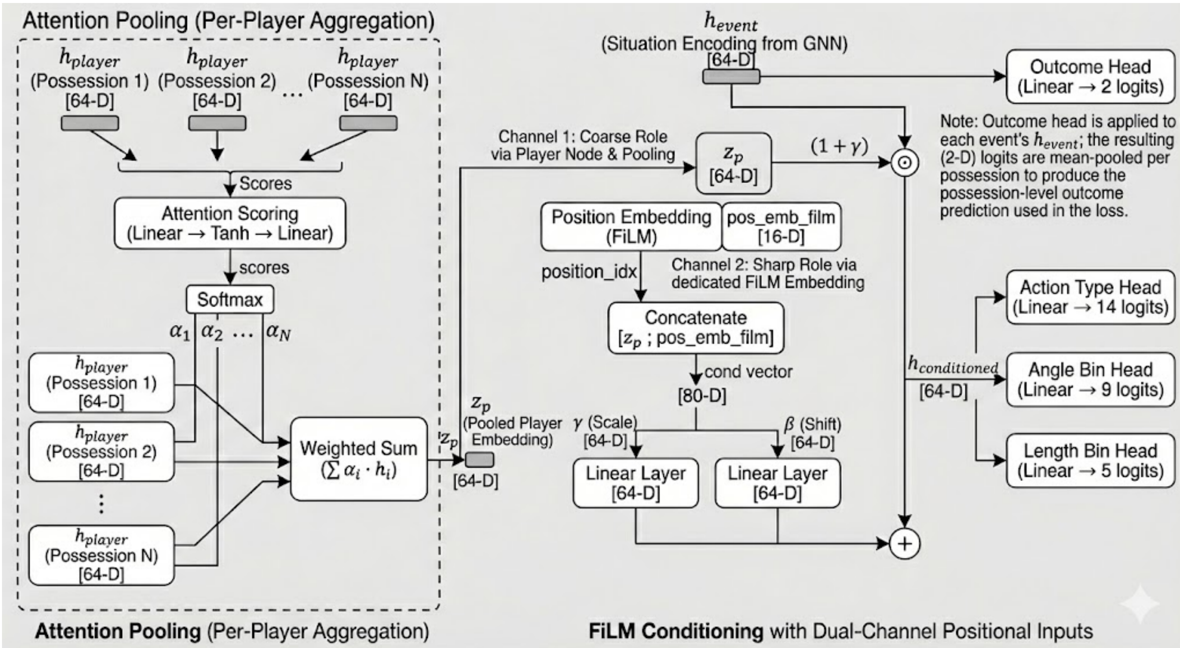


Figure 2: Full model architecture. *Left*: attention pooling aggregates per-possession player embeddings into a global trait z_p. *Right*: FiLM conditioning modulates the GNN's situation encoding with player-specific scale and shift. The outcome head operates on h^{event} alone, ensuring outcome prediction is player-independent.

6 Model Variation Components

We study eight modular components across 11 pipeline configurations (Table 1). Appendix D provides a complete hyperparameter specification for every variant.

C1. Split-context edges: separate teammate vs. opponent context relations.

C2. Position ablation: remove all explicit position information from the learned representation; FiLM receives z_p alone (64-D).

C3. Player-aware sampling: batch structure of K=16 players × M=6 possessions, with tuned contrastive preset.

C4. Uniformity pressure (t, λ_{uni}): Gaussian-potential sensitivity and weight; position-ablated variants use elevated settings (t=4.0, λ_{uni}=1.0).

C5. EMA alignment: cross-batch alignment loss with momentum memory bank (decay 0.999).

C6. Acts-in edge dropout: drops all actor→event edges with p=0.3 during training, forcing FiLM and z_p to carry more of the behavioral prediction burden.

C7. Position-group prediction (λ_{pos}): auxiliary loss predicting coarse position from z_p, reintroducing positional structure as a soft target.

C8. Group-weighted uniformity: 3× repulsive force for same-position-group pairs, sharpening within-group discrimination.

7 Evaluation Protocol

Each model is evaluated on 14 complementary metrics in six categories:

Table 1: Pipeline variants and active components. Blank = default/inactive.

Model	Split Ctx	Pos Abl	Player Samp	t	λ uni	EMA	λ align	Acts-In Drop	λ pos	Group Wt
baseline				2	.3					1
player_samp			✓	2	.5					1
split_ctx	✓			2	.3					1
split_ctx_ps	✓		✓	2	.5					1
pos_ablated		✓		2	.3					1
pos_abl_split_ctx	✓	✓		4	1.0					1
pos_abl_split_ctx_v2	✓	✓		4	.7					1
pos_abl_split_ctx_ema	✓	✓		4	1.0	✓	.3			1
pos_abl_split_ctx_ema_v2	✓	✓		4	1.0	✓	.1			1
acts_in_dropout	✓	✓		4	1.0			.3		1
acts_in_dropout_pos_gu	✓	✓		4	1.0			.3	.3	3

Model-based behavioral fidelity (wt. 0.28): Spearman ρ between embedding cosine distance and JS divergence of predicted action distributions (within 8 position$\times$gender groups); absolute top-k JS of the 10 closest neighbors; substitute quality ratio.

Empirical behavioral fidelity (wt. 0.09): non-self-referential check using observed actions bucketed by coarse game state (pitch third $\times$ under-pressure = 6 buckets, 5 action categories). EB Spearman ρ and EB substitute ratio.

Cross-context robustness (wt. 0.23): for 311 multi-competition players, competition-split self-consistency (mean rank and Hit@10).

Pseudo-ground-truth retrieval (wt. 0.12): a 15-pair heuristic probe set assembled from public analytics articles and media comparisons (see Appendix A). This is used only as a directional sanity check and should not be interpreted as expert-validated ground truth.

External validation (wt. 0.20): qualitative head-coach scoring (6 queries scored 1–5 by an LLM evaluator; the evaluator used was Claude Opus 4.6) and FIFA attribute comparison (16 non-GK pairs).

Supplementary (wt. 0.08): random-half self-consistency and held-out macro-F1.

Three heuristic baselines calibrate expectations: mean features, action-type histogram, and FIFA attributes.

8 Results

8.1 Held-out predictive validity

Table 2 reports held-out test performance for `acts_in_dropout_pos_gu`. Action-type accuracy reaches 0.839 (macro F1 0.713); outcome AUC exceeds 0.97. The F1 spread across non-collapsed models is small (0.050), with position-ablated variants paying a modest penalty for shifting capacity toward embedding geometry.

8.2 Retrieval and self-consistency

Table 3 presents pseudo-ground-truth retrieval and cross-context self-consistency side by side. The reported pseudo-GT mean ranks for the top models are relatively close, so they should not be interpreted as a clean statistical separation on their own. Player-sampling variants have conspicuously high median GT ranks (71–73) and the worst competition-split MR (>89), reflecting retrieval that does not generalize well across contexts. The `pos_abl_split_ctx` family is strongest on competition-split consistency (MR 67.3–67.7), with `acts_in_dropout` achieving the best competition-split Hit@10

Table 2: Held-out test metrics (`acts_in_dropout_pos_gu`).

Task	**Metric**	**Value**
Action Type (14-way)	Accuracy / Macro F1	0.839 / 0.713
Angle Bin (9-way)	Accuracy / Macro F1	0.674 / 0.609
Length Bin (5-way)	Accuracy / Macro F1	0.740 / 0.657
Outcome: Shot	AUC-ROC	0.975
Outcome: Goal	AUC-ROC	0.988

(0.351).

Table 3: Pseudo-GT retrieval (lower MR better; higher Hit better) and cross-context self-consistency.

	Pseudo-GT (15 pairs)				**Competition Split (311 players)**		
Model	MR	Med	H@10	H@50	Cos	MR	H@10
split_ctx	**56.1**	40	.233	.600	.859	83.9	.215
pos_abl_split_ctx_ema	62.6	37	.233	**.700**	.769	67.7	.318
pos_abl_split_ctx	63.1	39	**.267**	.633	.766	**67.3**	.346
acts_in_dropout_pos_gu	66.7	41	.233	.567	.746	74.0	.328
acts_in_dropout	68.2	32	**.267**	.633	.763	68.9	**.351**
baseline	67.6	**30**	.233	.667	.885	81.6	.238
player_samp	70.2	71	.200	.333	.569	93.5	.264
split_ctx_ps	71.4	73	.167	.333	.573	89.3	.262
pos_abl_split_ctx_v2	198.8	161	.067	.200	.325	200.8	.076

8.3 Behavioral fidelity: model-based and empirical

Table 4 consolidates model-based and empirical behavioral metrics. The highest model-based ρ (`pos_ablated`, 0.974) is partly consistent with embedding compression (mean cosine distance 0.085); among retrieval-viable models, `pos_abl_split_ctx_ema` leads (0.944). The EB rankings differ from the model-based rankings, suggesting that the EB view is not redundant with the model's own internal notion of similarity. `acts_in_dropout` leads EB substitute ratio (1.953), with `acts_in_dropout_pos_gu` second (1.745). Conversely, `pos_ablated`'s inflated model-based ρ does not carry over to EB ρ (0.196), indicating that rank-based compression effects are metric-specific.

8.4 External validation

The FIFA heuristic baseline leads FIFA-specific metrics because it directly optimizes that attribute space. Among the learned models, `acts_in_dropout_pos_gu` achieves the strongest external-validation profile shown here, with the best learned-model FIFA agreement (7.7) and the highest qualitative score (3.5/5). These results should still be interpreted cautiously: the qualitative judgments come from an LLM evaluator (Claude Opus 4.6), and the FIFA comparison is small (16 non-GK pairs). Accordingly, this section is best read as supportive evidence rather than definitive proof of tactical interchangeability.

8.5 Embedding structure and counterfactual validation

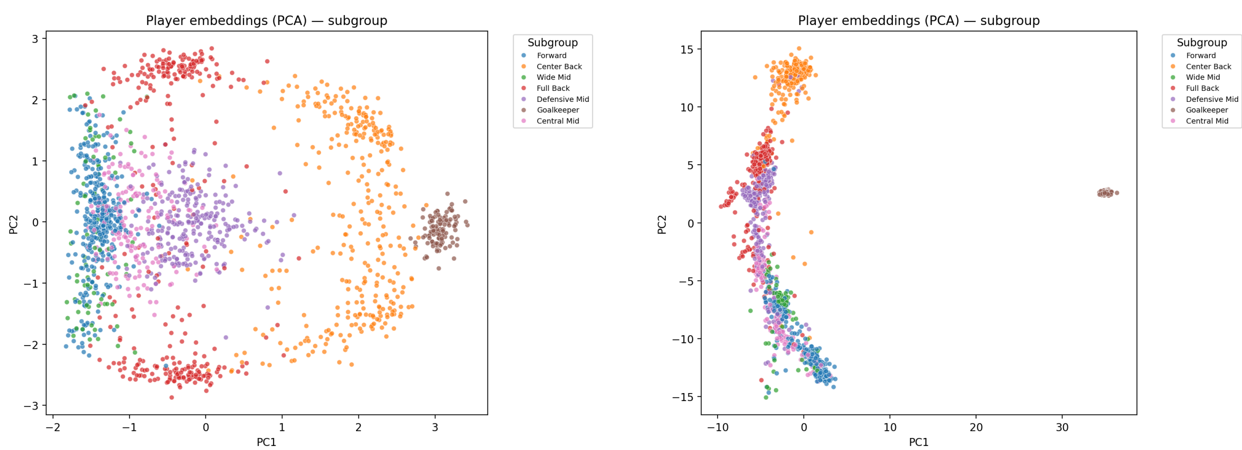
Trait-swapping validation confirms that the model's predictions change when different z_p are substituted into the same encoded state. A caveat is that, because actor–event edges are present at inference, some actor identity can leak into h^{event} before trait substitution. This therefore validates

Table 4: Behavioral fidelity. Model-based metrics use the model's own action predictions; EB metrics use observed player actions. Higher ρ and ratio are better; lower JS is better.

	Model-Based			Empirical Behavioral	
Model	ρ	Top-k JS	Ratio	EB ρ	EB Ratio
pos_ablated	**.974**	.00111	8.59	.196	1.66
pos_abl_split_ctx_ema	.944	.00104	7.95	.207	1.72
pos_abl_split_ctx	.929	.00141	6.63	.207	1.62
acts_in_dropout	.928	**.00103**	7.75	.207	**1.95**
acts_in_dropout_pos_gu	.922	.00105	7.62	.195	1.75
player_samp	.788	.00112	11.91	.223	1.56
split_ctx_ps	.778	**.00103**	**13.11**	**.230**	1.70
baseline	.744	.00330	3.84	.188	1.32
split_ctx	.710	.00461	3.13	.189	1.25

Table 5: External validation. FIFA: 16 non-GK query-neighbor pairs. Qualitative: 6 queries scored 1–5 by Claude Opus 4.6 (see Appendix C).

Model	FIFA M-6	FIFA Pos%	Qual. Total	Qual. Avg
acts_in_dropout_pos_gu	**7.7**	50	**21**/30	**3.5**
acts_in_dropout	7.9	50	18/30	3.0
pos_abl_split_ctx_ema	9.1	56	18/30	3.0
pos_abl_split_ctx	9.0	50	17/30	2.8
baseline	8.6	44	13/30	2.2
split_ctx	8.5	56	11/30	1.8
h_fifa_attributes	**5.0**	**75**	—	—



(a) Selected model embedding PCA: position subgroups remain visually separated despite removing positional input from FiLM.

(b) Baseline embedding PCA: position subgroup clustering is also visible, with tighter concentration in some groups.

Figure 3: PCA projection of global player embeddings by position subgroup.

FiLM sensitivity rather than a perfectly clean counterfactual intervention.

8.6 Qualitative neighbor inspection

Table 6 reports top-5 neighbors for selected query players, providing face-validity checks. Ronaldo retrieves clinical center forwards; Messi retrieves creative playmaker-forwards; De Bruyne retrieves progressive midfield creators; Bronze retrieves elite attacking right-backs from women's football. Gender-aware retrieval restricts Bronze's gallery to female players, yet her neighbors are stylistically meaningful. Face-validity for famous players is necessary but not sufficient; evaluating lesser-known or emerging players would require domain-expert judgments.

Table 6: Top-5 nearest neighbors (`acts_in_dropout_pos_gu`).

#	Query	Top-5 Neighbors	Cos
Ronaldo	Cristiano Ronaldo (CF)	Morata	.954
		Lautaro Martínez	.949
		Memphis Depay	.945
		V. Boniface	.940
		Kane	.935
Messi	Lionel Messi (RCF)	Griezmann	.939
		Shaqiri	.915
		Arda Güler	.907
		Kramarić	.893
		Mertens	.891
KDB	Kevin De Bruyne (CAM)	A. Ramsey	.911
		Zieliński	.901
		Bruno Fernandes	.893
		Griezmann	.889
		Miranchuk	.885
Bronze	Lucy Bronze (RB)	Lynn Wilms	.973
		Giulia Gwinn	.966
		Emily Fox	.965
		O. Hernández	.963
		E. Carpenter	.956

9 Weighted Model Selection

No single model dominates all metrics. To integrate trade-offs, we use a weighted composite with min-max normalization across the 11 GNN models:

$$S_m = \frac{\sum_i w_i \cdot \text{norm}_{m,i}}{\sum_i w_i \cdot \mathbf{1}[\text{available}]}.$$

Weight justification and the full raw-metric table appear in Appendix B.

Under this weighting scheme, `acts_in_dropout_pos_gu` achieves the highest composite score (0.867), followed very closely by `acts_in_dropout` (0.864). Across four sensitivity analyses (balanced, behavioral-heavy, retrieval-heavy, and external-heavy), the two acts-in-dropout variants remain the top two models, suggesting that the high-level selection is moderately robust to reasonable weight changes.

Variant	#1	#2	#3
Balanced (default)	**Selected model** (.867)	Acts-in dropout (.864)	EMA-v2 (.860)
Behavioral-heavy	Acts-in dropout (.862)	**Selected model** (.857)	`pos_ablated` (.850)
Retrieval-heavy	Acts-in dropout (.877)	**Selected model** (.874)	EMA (.855)
External-heavy	**Selected model** (.878)	Acts-in dropout (.859)	`pos_ablated` (.834)

Caveats. The composite score is a model-selection aid, not a statistical proof of superiority. It depends on manually chosen weights, partial metric availability for some models, and evaluation protocols of uneven credibility. For example, `pos_ablated` is missing qualitative data, while `pos_abl_split_ctx_ema_v2` is missing both qualitative and EB results. Accordingly, the composite ranking should be interpreted as a transparent aggregation of available evidence rather than a definitive ordering.

10 Discussion

Progressive design insights. The 11 pipelines reflect a three-stage exploration. *Stage 1 (baselines)*: `split_ctx` achieves the best pseudo-GT mean rank (56.1) but weak behavioral fidelity and poor qualitative results, suggesting that positional structure remains too dominant in the embedding. *Stage 2 (position ablation)*: removing explicit positional input pushes the representation toward behavior, and the `pos_abl_split_ctx` family achieves the strongest competition-split consistency among the tested variants. In the present sweep, lowering the stronger ablation setting from $\lambda_{\text{uni}} = 1.0$ to 0.7 is associated with severe degradation, but this should be interpreted as evidence about the tested configuration grid rather than a universal threshold. *Stage 3 (acts-in dropout)*: dropping actor edges during training further increases reliance on the trait pathway. The EB analysis supports the acts-in-dropout family as the strongest observed-behavior match among the evaluated learned models, with group-weighted uniformity contributing more to qualitative coherence than to clear EB separation.

Key tensions. (1) *Player sampling vs. substitute discovery*: `player_samp` performs very strongly on self-consistency but poorly on cross-context substitute retrieval, indicating that "same player = close" is not the same objective as "different players with similar function = close". (2) *Behavioral fidelity vs. retrieval*: `pos_ablated` attains the highest model-based ρ but only middling retrieval quality, suggesting that compressed distance structure can inflate rank-based agreement. (3) *Pseudo-GT mean rank vs. qualitative quality*: `split_ctx` has the best pseudo-GT mean rank yet the weakest qualitative score among the displayed learned models, showing that these evaluation views are not interchangeable.

Role adjacency vs. tactical equivalence. The model captures similarity in on-ball decision distributions, but this does not always imply tactical interchangeability. Rúben Dias for Van Dijk is structurally coherent: both are elite ball-playing centre-backs who organize the line, step out, and distribute progressively, even though Dias is more aggressive and front-footed while Van Dijk favours calmer, longer distribution. Bruno Fernandes for De Bruyne is more role-adjacent than fully equivalent: both are creative midfielders who select similar action types (through-balls, crosses, shots) in advanced zones, but De Bruyne operates as a tempo-setting, half-space progression specialist within a controlled positional structure, whereas Bruno is more direct and transition-heavy. A 14-way action-type prediction cannot resolve such mechanistic distinctions; these would require modelling passing rhythm, off-ball positioning, and risk calibration that the current feature set does not capture. This suggests the model is better at capturing *what* decisions a player tends to make than fully capturing *how* those decisions are executed, which matters for tactical scouting.

Limitations.

- **Evaluation credibility**: external validation relies on LLM-compiled pairs, LLM-scored judgments, and a small FIFA comparison (16 pairs). Human expert evaluation would substantially strengthen the evidence.
- **Partially self-referential behavioral metric**: model-based ρ and JS measure alignment with the model's own predictions. The EB metric partially addresses this, but its coverage (5 action categories, 6 game-state buckets, nine players in the EB substitute-quality protocol) is limited relative to the model's fine-grained conditional policies.
- **Pseudo-ground-truth provenance**: the probe-pair appendix mixes sources of uneven rigor, and several entries are supported only by broad public comparisons rather than exact expert-validated similarity labels.
- **No strong learned baselines**: comparisons against a sequential encoder or MLP over possession summaries would clarify the graph formulation's contribution.
- **Weight-dependent selection**: the composite ranking depends on manually chosen weights.
- **Temporal stability**: fixed data window; embeddings may not track evolving styles.
- **Low-data players**: the 50-possession minimum excludes many real scouting targets.
- **Retrieval scale**: galleries contain roughly 1,600 players; production use would require much larger galleries.
- **Counterfactual leakage**: actor–event edges allow actor identity to bleed into h^{event} before trait substitution.

11 Conclusion

We define player similarity as similarity of conditional decision policies and propose a pipeline that learns behaviorally grounded embeddings through heterogeneous graph encoding, attention-based trait extraction, and FiLM conditioning. Across 11 ablation variants and a broad evaluation suite, the paper supports four restrained conclusions:

1. **Position ablation plus split-context edges is beneficial in the tested setup.** Within the evaluated grid, this combination gives the strongest non-collapsed cross-context behavior.
2. **Player-aware sampling appears misaligned with substitute retrieval.** It improves self-re-identification more than cross-player functional similarity.
3. **Acts-in dropout is a strong ingredient in the best-performing family.** The EB protocol supports this family as producing better observed behavioral matches than the weaker learned alternatives shown here, though the protocol is not granular enough for clean causal isolation.
4. **The selected model is best interpreted as a weighted best-compromise model.** `acts_in_dropout_pos_gu` ranks first under the chosen composite, but it does not dominate every metric and should therefore be presented as the most balanced tested configuration, not the universally best model.

The paper's main methodological contribution is promising, but the empirical claims should remain calibrated to the strength of the current validation evidence.

Code Availability

The full code used for preprocessing, graph construction, model training, evaluation, and figure generation is available in the accompanying repository.

Appendix

A Pseudo Ground-Truth Pairs

The 15 pairs below form a *heuristic probe set* rather than a formal ground-truth benchmark. They were assembled from a mixture of public analytics articles, recruitment pieces, public similarity tools, and analyst-style media comparisons. The purpose of this set is only to test whether the learned space retrieves obviously unreasonable neighbors or broadly plausible stylistic alternatives.

Because the underlying sources are heterogeneous and not all are equally rigorous, disagreement with this table should not automatically be interpreted as model failure. For that reason, the main paper treats pseudo-GT results as directional sanity checks rather than definitive retrieval supervision.

Selection criteria. The probe pairs were chosen to span multiple position groups, include both men's and women's football, and reflect commonly discussed stylistic comparisons in public discourse. They are intentionally heterogeneous and are therefore not suitable for fine-grained statistical claims. Their value is diagnostic: they help identify retrieval outputs that are clearly implausible, but they do not provide a gold-standard ranking target.

Table 7: Pseudo ground-truth similar-player pairs.

Tier	Player A	Player B	Positions	Poss A	Poss B	Heuristic Rationale
1	Modrić	Kroos	RCM / LDM	828	584	Tempo-controlling deep playmakers (Real Madrid partnership).
1	Van Dijk	R. Dias	LCB / LCB	562	618	Dominant ball-playing centre-backs; common public comparison.
1	J. Alba	Robertson	LB / LWB	570	255	Widely discussed attacking fullback comparison from recruitment-style analysis.
1	TAA	Hakimi	RDM / RB	87	333	Attack-first right-sided creators; included despite TAA's low possession count.
1	Bonmatí	Putellas	RCM / LCM	854	465	High-level women's midfield comparison with overlapping creative profiles.
2	Saka	Dembélé	RW / RW	518	407	Inverted-winger archetype comparison.
2	Kane	Lewandowski	CF / LCF	676	335	Clinical #9s with strong finishing and link play.
2	Wirtz	De Bruyne	LAM / CAM	1820	487	Publicly framed as advanced creators with similar attacking influence.
2	Renard	Bright	LCB / RCB	479	639	Widely recognized top-level women's centre-back comparison.
2	Hemp	Caldentey	LW / LW	787	840	Creative left-sided forwards with hybrid scoring and chance-creation profiles.
2	Kroos	Enzo F.	LDM / CDM	584	351	Deep midfield replacement-style comparison.
2	Bellingham	Griezmann	CAM / CAM	607	690	Goal-threat attacking-midfield comparison; noisier than tier-1 pairs.
3	Musiala	Foden	LW / LW	442	477	Common stylistic comparison, but role nuance differs.
3	Yamal	N. Williams	RW / LW	252	317	Public comparison tied partly to shared tournament context rather than same-side role.
3	Neuer	Donnarumma	GK / GK	339	296	Distribution-oriented elite goalkeepers; stylistic details still differ.

B Full Raw Metrics and Weights

Table 8: Evaluation metrics with category weights.

#	Metric	Wt	Category (total)
1	Spearman ρ (model-based)	.14	Model-Based Beh. (0.28)
2	Absolute Top-k JS (model-based)	.10	
3	Sub Quality Ratio (model-based)	.04	
4	EB Spearman ρ	.07	Empirical Beh. (0.09)
5	EB Substitute Ratio	.02	
6	Comp-Split Mean Rank	.17	Cross-Context (0.23)
7	Comp-Split Hit@10	.06	
8	Random-Half Mean Rank	.05	Supplementary (0.08)
9	Test F1 Avg	.03	
10	Pseudo-GT Mean Rank	.08	Pseudo-GT (0.12)
11	Pseudo-GT Hit@50	.04	
12	FIFA Main-6 Diff	.05	External (0.20)
13	FIFA Position Match%	.05	
14	Qualitative Avg	.10	

Table 9: Raw metric values for all 11 GNN models. Direction: $\rho \uparrow$, JS $\downarrow$, MR $\downarrow$, H@K $\uparrow$, Ratio $\uparrow$, FIFA $\downarrow$, Pos% $\uparrow$, F1 $\uparrow$, Qual $\uparrow$, EB $\rho \uparrow$, EB Ratio $\uparrow$. "—" = unavailable.

Model	ρ	JS	Comp MR	C.H@10	Rnd MR	GT MR	GT H@50	Ratio	FIFA	Pos%	F1	Qual	EB ρ	EB Rat	Score
acts_in_dropout_pos_gu	.922	.00105	74.0	.328	8.2	66.7	.567	7.62	7.7	50	.660	3.5	.195	1.75	**.867**
acts_in_dropout	.928	.00103	68.9	.351	8.7	68.2	.633	7.75	7.9	50	.657	3.0	.207	1.95	.864
pos_abl_split_ctx_ema_v2	.934	.00145	67.5	.349	6.1	64.3	.633	6.35	8.8	50	.674	—	—	—	.860†
pos_ablated	.974	.00111	81.3	.275	13.9	69.6	.567	8.59	8.6	50	.687	—	.196	1.66	.847†
pos_abl_split_ctx_ema	.944	.00104	67.7	.318	9.3	62.6	.700	7.95	9.1	56	.650	3.0	.207	1.72	.846
pos_abl_split_ctx	.929	.00141	67.3	.346	6.3	63.1	.633	6.63	9.0	50	.675	2.8	.207	1.62	.820
player_samp	.788	.00112	93.5	.264	2.9	70.2	.333	11.91	8.9	50	.700	—	.223	1.56	.735†
split_ctx_ps	.778	.00103	89.3	.262	2.7	71.4	.333	13.11	9.2	25	.688	—	.230	1.70	.692†
baseline	.744	.00330	81.6	.238	26.1	67.6	.667	3.84	8.6	44	.692	2.2	.188	1.32	.593
split_ctx	.710	.00461	83.9	.215	36.4	56.1	.600	3.13	8.5	56	.684	1.8	.189	1.25	.536
pos_abl_split_ctx_v2	.686	.00696	200.8	.076	177.8	198.8	.200	3.07	9.8	38	.639	—	.104	1.21	.023†

† Scored on fewer than 14 metrics; weight renormalized over available dimensions.

C Per-Query Qualitative Scoring

Six famous-player queries were scored by an LLM evaluator (Claude Opus 4.6) on a 1–5 scale for face-validity of top-5 neighbors. The evaluator was given the query player's name, position, and broad playing style, together with the top-5 retrieved neighbors and cosine similarities, and asked whether the neighbors were stylistically plausible substitutes (5 = excellent match, 1 = poor or wrong position group). These scores are heuristic and directional, not authoritative; inter-rater reliability with human experts has not been assessed.

Table 10: Per-query qualitative scores (1–5). M1=baseline, M2=`pos_abl_split_ctx`, M3=`split_ctx`, M4=`pos_abl_split_ctx_ema`, M5=`acts_in_dropout`, M6=`acts_in_dropout_pos_gu`.

Query	M1	M2	M3	M4	M5	M6
Neuer (GK)	2	3	3	**4**	3	**4**
Van Dijk (CB)	3	2	3	2	3	**4**
TAA (RDM)	**3**	2	1	2	2	2
De Bruyne (CAM)	3	3	2	3	3	**4**
Messi (RCF)	1	**3**	1	**3**	**3**	**3**
Mbappé (LW)	1	**4**	1	**4**	**4**	**4**
Total	13	17	11	18	18	**21**

Key patterns. Position-ablated models (M2, M4–M6) outperform the more position-driven models (M1, M3) on creative or unusual players such as Messi and Mbappé. `split_ctx` tends to retrieve positional peers rather than stylistic matches. The selected model retrieves Rúben Dias for Van Dijk and Bruno Fernandes for De Bruyne, which helps explain its higher qualitative average. No model scores 5 on any query, and TAA remains difficult across all variants.

D Model Variation Details

Each pipeline variant is identified by a shorthand name that concatenates its active components. Table 11 provides the full specification.

Table 11: Complete hyperparameter specification for every pipeline variant. Defaults (applied when a cell is blank): split-context = off, position ablation = off, player sampling = off, t=2.0, λ_{uni}=0.3, EMA = off, acts-in dropout p=0, λ_{pos}=0, group weight w_g=1.

Name	Split Ctx	Pos Abl	Plyr Samp	t	λ_{uni}	EMA	λ_{align}	Drop p	λ_{pos}	w_g	Motivation
`baseline`				2	.3					1	Default configuration; reference point.
`player_samp`			✓	2	.5					1	Test whether structuring batches by player helps.
`split_ctx`	✓			2	.3					1	Separate teammate/opponent context edges.
`split_ctx_ps`	✓		✓	2	.5					1	Combine split-context with player sampling.
`pos_ablated`		✓		2	.3					1	Remove position from FiLM input.
`pos_abl_split_ctx`	✓	✓		4	1.0					1	Ablation + split-context + strong uniformity.
`pos_abl_split_ctx_v2`	✓	✓		4	0.7					1	Reduced uniformity (λ=0.7); stress-test for collapse sensitivity.
`pos_abl_split_ctx_ema`	✓	✓		4	1.0	✓	.3			1	Add EMA cross-batch alignment.
`pos_abl_split_ctx_ema_v2`	✓	✓		4	1.0	✓	.1			1	Lower alignment weight; archived comparison variant.
`acts_in_dropout`	✓	✓		4	1.0			.3		1	Drop actor edges during training to force stronger reliance on the trait pathway.
`acts_in_dropout_pos_gu`	✓	✓		4	1.0			.3	.3	3	Add position-group auxiliary loss and stronger within-group repulsion for improved qualitative coherence.

E FIFA Radar Diagram

Figure 4 provides an external sanity check by comparing the FIFA/EA FC attribute profiles of each query player and the model's top-1 retrieved substitute. The six outfield attributes (Pace, Shooting, Passing, Dribbling, Defending, Physical) are composite ratings on a 0–100 scale compiled by EA Sports from scouting networks; goalkeeper pairs instead use Diving, Handling, Kicking, Positioning, and Reflexes. Because the GNN is trained exclusively on StatsBomb on-ball decisions and never sees FIFA data, agreement between the radar profiles is a genuine out-of-distribution validation signal. Across the 16 non-GK pairs evaluated for ACTS_IN_DROPOUT_POS_GU, the mean main-6 absolute difference is 7.7 (Table 5), substantially closer than a random same-position baseline but still above the FIFA heuristic (5.0), which optimises directly on that attribute space.

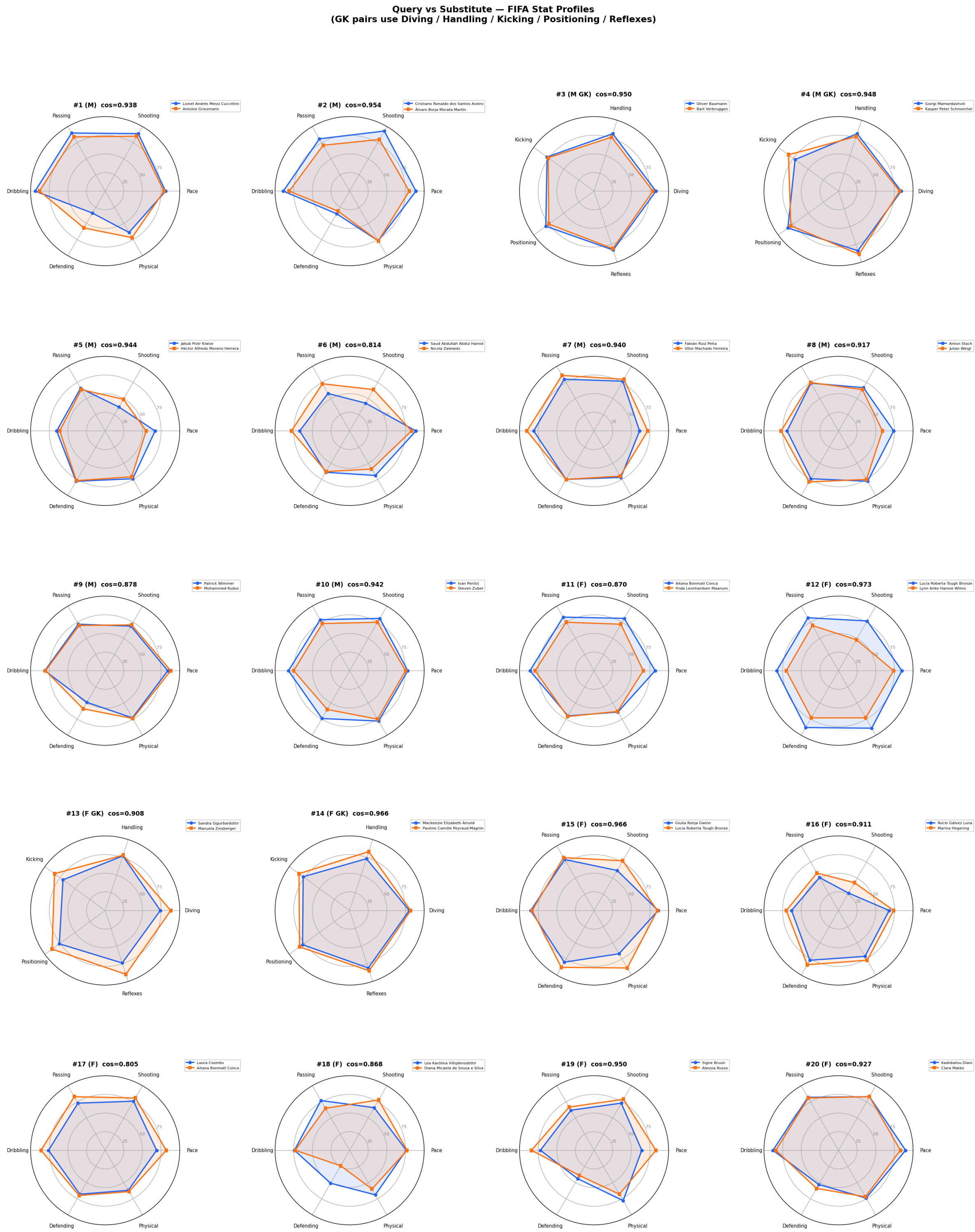


Figure 4: FIFA attribute radar profiles for query–substitute pairs (ACTS_IN_DROPOUT_POS_GU). Blue = query, orange = top-1 cosine substitute. Outfield pairs use six main attributes; goalkeeper pairs use five GK-specific axes. Close polygon overlap indicates agreement between GNN behavioral similarity and external FIFA attributes.

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